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FORMS

OVERVIEW

- **How forms work**
- **The `form` element**
- **Text entry controls**
- **Buttons**
- **Menus**
- **Specialized inputs**
- **Form accessibility**
- **Form design tips**

How Forms Work

Web forms have two components:

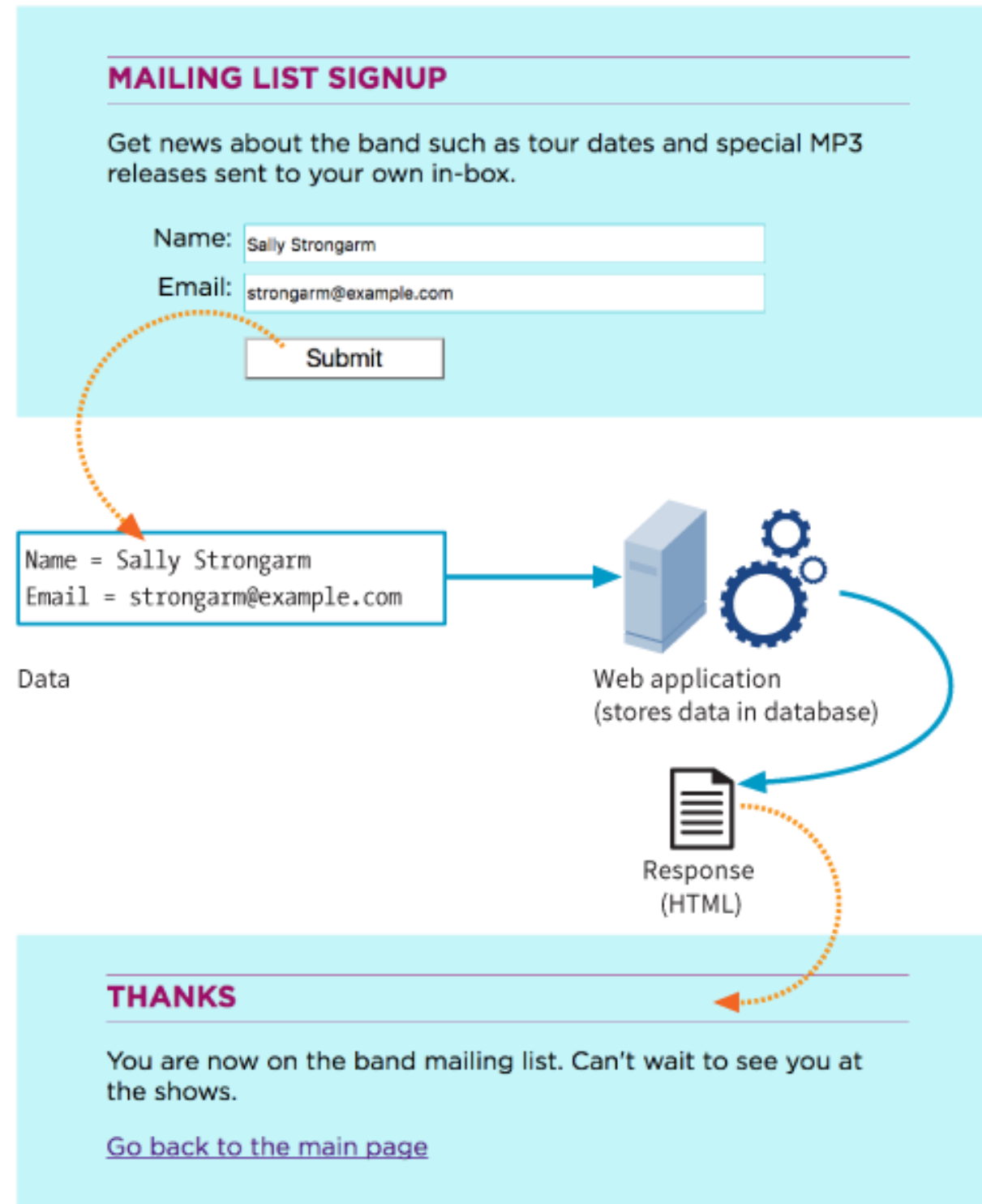
- The **form on the page** that collects input
- An **application on the server** that processes the collected information

Web Form Transaction

1. Browser renders the form inputs as indicated in the markup.

2. User enters information in the form and hits Submit.

3. The browser encodes the information entered and sends it to the server.



4. The application processes the information.

5. The application returns a response (for example, a thank you message or reloading the page).

Web Processing Applications

Web forms may be processed by any of the following technologies:

- PHP (*.php*)
- Microsoft ASP (*.asp*)
- Microsoft ASP.net (*.aspx*)
- Ruby on Rails
- JavaServer Pages (*.jsp*)
- Python

The form Element

```
<form action="URL" method="POST or GET">  
  <!-- Form content and inputs here -->  
</form>
```

- The **form** element is a container for all the content in the form and its input controls.
- The **action** and **method** attributes are necessary for interacting with the processing program.

The action Attribute

```
<form action="mailinglist.php" method="POST">
```

The **action** attribute provides the location of the script or application that will process the collected form data.

The method Attribute

```
<form action="mailinglist.php" method="POST">
```

The **method** attribute specifies how the encoded information should be sent to the server (GET or POST):

- **GET**: The encoded data is tacked onto the URL sent to the server:
- **POST**: Data is send in a separate transaction and can be encrypted with HTTPS.

NOTE: POST is the most common method.

Form Control Elements

Form control elements (also called **widgets**) collect data from the user. A few examples:

Type	Description
<code><input type="text"></code>	Displays a single-line text input field
<code><input type="radio"></code>	Displays a radio button (for selecting one of many choices)
<code><input type="checkbox"></code>	Displays a checkbox (for selecting zero or more of many choices)
<code><input type="submit"></code>	Displays a submit button (for submitting the form)
<code><input type="button"></code>	Displays a clickable button

Form Control Elements

First Name:

Last Name:

- ☒ Punk rock
- ☒ Indie rock
- ☐ Hip Hop
- ☐ Rockabilly

80s band?

Form Control Elements (cont'd)

Form controls collect data in **variable/value pairs**.

Examples:

variable = "email"

value = jen@example.com

variable = "color"

value = green

Variables (the name Attribute)

- A **variable** is a bit of information collected by a form control (example: **the user's last name**).
- The required **name** attribute in the control element provides the name of the variable for that control:

```
<input name="lastname">
```

NOTE: The variable name is also programmed into the web processing script or app, so the name in the markup must match the name in the processor.

Values

- The data entered or selected by the user in the form control is the **value** that gets sent to the server. It is paired with the variable for that control.
- You can provide a default value with the **value** attribute:

Name: `<input name="lastname" value="Unknown">`

In this example, if the text input is left blank, the value “Unknown” would be sent to the server for the variable “lastname”.

FORM CONTROL ELEMENTS

Text Entry Input

`<input type="text">`

Favorite color:

```
<input type="text" name="color" value="Red" maxlength="24">
```

type: Type of input control, in this case a single-line text field

name: Required variable name

value: Default text that appears in the field and is sent to server if the field is left blank

maxlength, minlength: Sets a character limit for the field

size: The length of the field in number of characters (CSS is more common for sizing)

FORM CONTROL ELEMENTS

Password Field

```
<input type="password">
```



```
<input type="password" name="pswd" maxlength="10">
```

- Like a text entry field, except the characters are obscured from view
- The data entered is *not* encrypted on the way to the server (unless it uses HTTPS, a secure web protocol).

FORM CONTROL ELEMENTS

Multi-line Text Entry

`<textarea> </textarea>`

Official contest entry:

Tell us why you love the band. Five winners will get backstage passes!

The band is totally awesome!

```
<textarea name="entry" rows="6" cols="64">This band is totally  
awesome!</textarea>
```

The content of the **textarea** element is the default value.

rows: The number of rows tall the field is initially drawn (users can write more)

cols: Width of initial text field, in number of characters

maxlength, minlength: Limits the number of characters that can be entered

Specialized Text Entry Fields

```
<input type="search">
```

```
<input type="email">
```

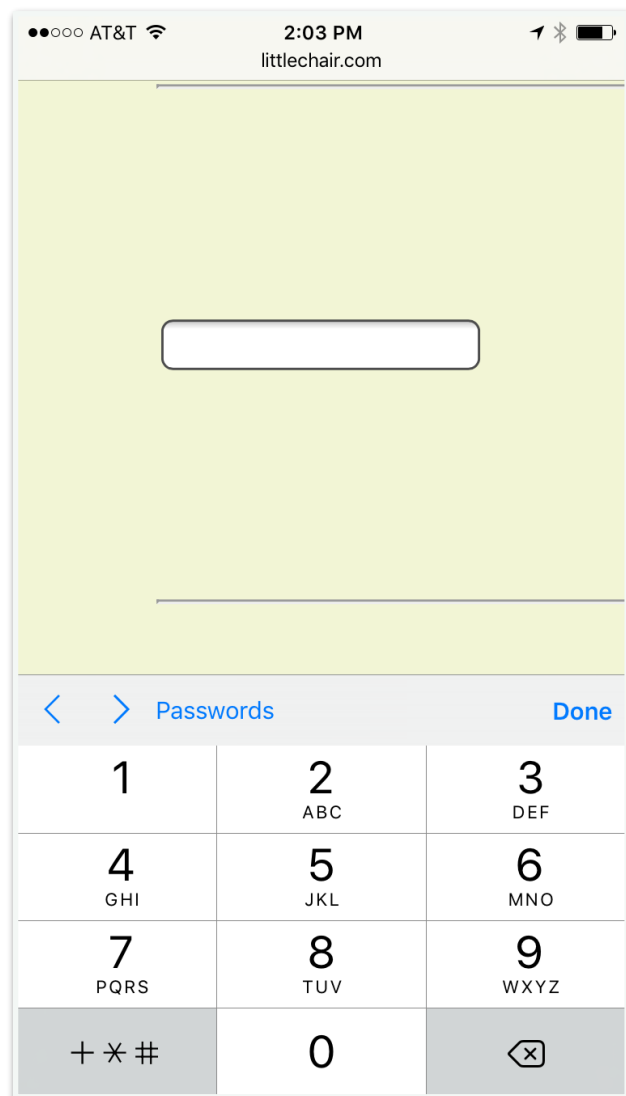
```
<input type="tel">
```

```
<input type="url">
```

- These input types are more semantically rich than a default text field.
- Browsers may provide keyboards tailored to the input type.
- Browsers may validate entries on the fly without using the server application.

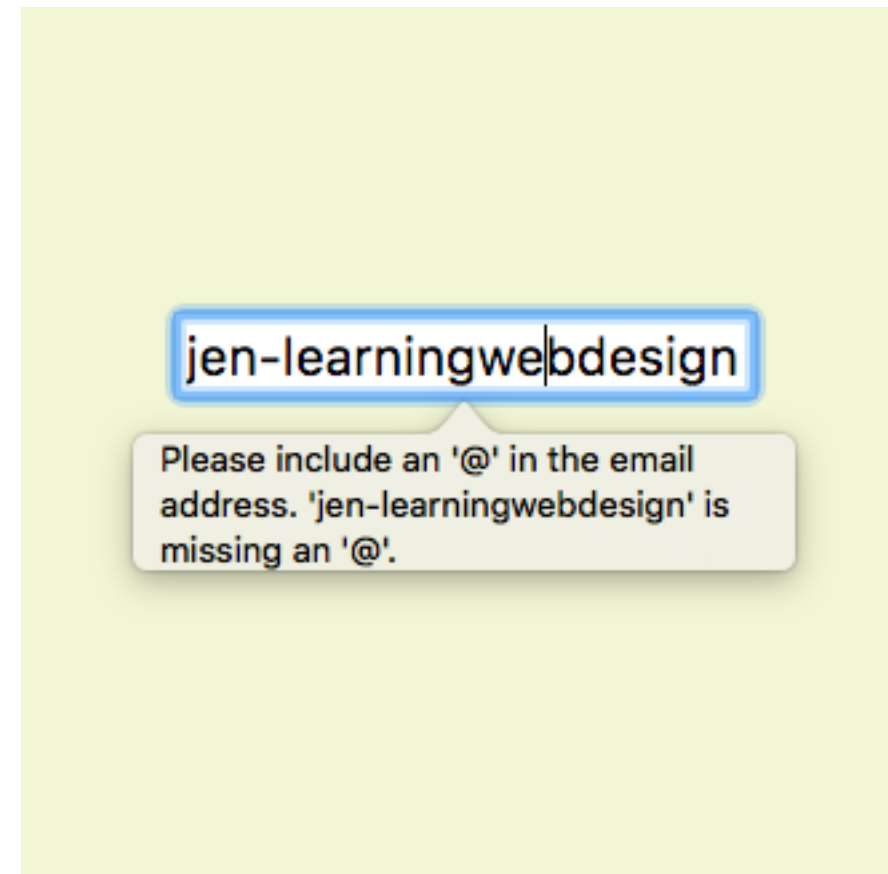
Specialized Text Entries (cont'd)

`<input type="tel" name="">`



Numerical keyboard provided on iOS

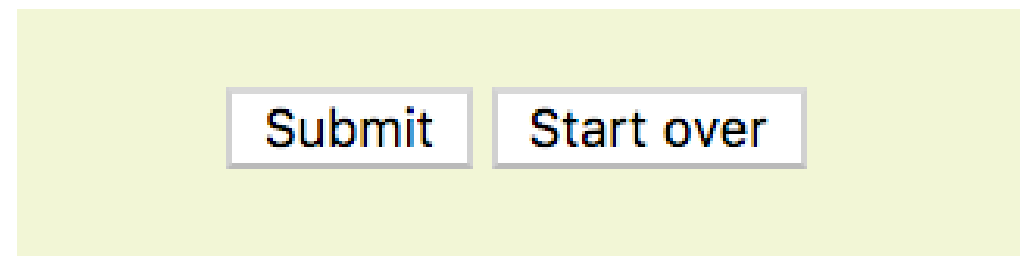
`<input type="email" name="">`



Opera looks for email address structure

FORM CONTROL ELEMENTS

Submit and Reset Buttons



```
<input type="submit">
```

- **Submits** the collected form data to the server. Does not require a variable name (**name** attribute):

```
<input type="reset" value="Start over">
```

- **Resets** the form to its defaults
- Less common with the rise of JavaScript for form handling
- Change the button text with the **value** attribute.

Custom Buttons

`<button> </button>`

The **button** element is used for creating custom buttons with JavaScript.

`<input type="button">`

Creates a custom button that has no predefined function and can be customized with JavaScript

`<input type="image" alt="">`

Allows an image to be used as a button to replace the Submit button. It requires a descriptive **alt** attribute value.

FORM CONTROL ELEMENTS

Radio Buttons

`<input type="radio">`

Only one radio button may be selected at a time.

How old are you?

- ☒ under 24
- ☐ 25 to 34
- ☐ 35 to 44
- ☐ 45+

```
<p>How old are you?</p>
```

```
<ol>
```

```
<li><input type="radio" name="age" value="under24" checked> under 24</li>
```

```
<li><input type="radio" name="age" value="25-34"> 25 to 34</li>
```

```
<li><input type="radio" name="age" value="35-44"> 35 to 44</li>
```

```
<li><input type="radio" name="age" value="over45"> 45+</li>
```

```
</ol>
```

NOTE: You can't belong to more than one age group, so radio buttons are the right choice for this list.

Radio Buttons (cont'd.)

```
<input type="radio" value="">
```

- Applying the same variable name to input elements binds them together as a mutually exclusive set of options.
- The value for each button must be provided with the **value** attribute.
- The **checked** attribute causes an option to be selected when the page loads. Only one input may be checked.

FORM CONTROL ELEMENTS

Checkbox Buttons

`<input type="checkbox">`

Multiple checkbox buttons
may be selected.

What type of music do you listen to?

- ☒ Punk rock
- ☒ Indie rock
- ☐ Hip Hop
- ☐ Rockabilly

```
<p>What type of music do you listen to?</p>
```

```
<ul>
```

```
  <li><input type="checkbox" name="genre" value="punk" checked> Punk rock</li>
```

```
  <li><input type="checkbox" name="genre" value="indie" checked> Indie rock</li>
```

```
  <li><input type="checkbox" name="genre" value="hiphop"> Hip Hop</li>
```

```
  <li><input type="checkbox" name="genre" value="rockabilly"> Rockabilly</li>
```

```
</ul>
```

NOTE: You can like more than one type of music, so checkbox buttons are the right choice for this list.

Checkbox Buttons (cont'd)

```
<input type="checkbox" value="">
```

- Applying the same variable name to input elements binds them together as a group.
- The value for each button must be provided with the **value** attribute.
- The **checked** attribute causes an option to be selected when the page loads. Multiple buttons in a group may be checked.

End of Lecture